of the micropyle with respect to funicle and chalaza. Most important ovule types are orthotropous, anatropous, hemianatropous and campylotropous. The types of ovule is given in Figure 1.8.

Orthotropous: In this type of ovule, the micropyle is at the distal end and the micropyle, the funicle and the chalaza lie in one straight verticalline. Examples: Piperaceae, Polygonaceae

Anatropous: The body of the ovule becomes completely inverted so that the micropyle and funiculus come to lie very close to each other. This is the common type of ovules found in dicots and monocots.

Hemianatropous: In this, the body of the ovule is placed transversely and at right angles to the funicle. Example: Primulaceae.

Campylotropous: The body of the ovule at the micropylar end is curved and more or less bean shaped. The embryo sac is slightly curved. All the three, hilum, micropyle and chalaza are adjacent to one another, with the micropyle oriented towards the placenta. Example: Leguminosae

In addition to the above main types there are two more types of ovules they are,

Amphitropous: The distance between hilum and chalaza is less. The curvature of the ovule leads to horse-shoe shaped nucellus. Example: some Alismataceae.

Circinotropous: Funiculus is very long and surrounds the ovule. Example: Cactaceae

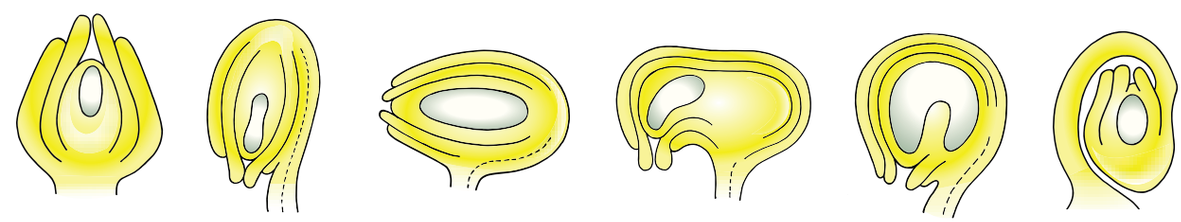
Megasporogenesis

The process of development of a megaspore from a megaspore mother cell is called **megasporogenesis**.

As the ovule develops, a single hypodermal cell in the nucellus becomes enlarged and functions as **archesporium**. In some plants, the archesporial cell may directly function as megaspore mother cell. In others, it may undergo a transverse division to form outer primary parietal cell and inner primary sporogenous cell. The parietal cell may remain undivided or divide by few periclinal and anticlinal divisions to embed the primary sporogenous cell deep into the nucellus. The primary sporogenous cell functions as a megaspore mother cell. The megaspore mother cell (MMO) undergoes meiotic division to form four haploid megaspores. Based on the number of megaspores that develop into the Embryo sac, we have three basic types of development: **monosporic, bisporic** and **tetrasporic**. The megaspores are usually arranged in a linear tetrad. Of the four megaspores formed, usually the chalazal one is functional and other three megaspores degenerate. The functional megaspore forms the female gametophyte or embryo sac. This type of development is called **monosporic** development (Example: *Polygonum*). Of the four megaspores formed if two are involved in Embryo sac formation the development is called **bisporic** (Example: *Allium*). If all the four megaspores are involved in Embryo sac formation the development is called **tetrasporic** (Example: *Peperomia*). An ovule generally has a single embryo sac. The development of monosporic embryo sac (*Polygonum* type) is given in Figure 1.9.

Development of Monosporic embryo sac.

To describe the stages in embryo sac development and organization the simplest monosporic type of development is given below.



(a) Orthotropous (b) Anatropous (c) Hemianatropous (d) Campylotropous (e) Amphitropous (f) Circinotropous

Figure 1.8 Types of ovule

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